

Exercise = 9.2

① Approximate $\sqrt{299^2 + 399^2}$ by means of differentials.

Sol:

formula: $\sqrt{299^2 + 399^2}$

Most approach value of given function

$$\text{is } 299 \approx 300 = x$$

$$399 \approx 400 = y$$

Mean of differentials

$$f(x, y) + \Delta f$$

$$\sqrt{x^2 + y^2} + \Delta f$$

$$= \text{partial derivatives } \Delta f = f_x \cdot dx + f_y \cdot dy$$

$$f_x = \frac{\partial f}{\partial x} ; f_y = \frac{\partial f}{\partial y}$$

Sol:

According to given statement

let $f(x, y) = \sqrt{x^2 + y^2}$ ——— ①
nearest value of "x" and "y"
w.r.t given statement

$$x = 300$$

$$; y = 400$$

$$x + dx = 299$$

$$; y + dy = 399$$

$$300 + dx = 299$$

$$; 400 + dy = 399$$

$$dx = 299 - 300$$

$$; dy = 399 - 400$$

$$dx = -1$$

$$; dy = -1$$

partial diff ① w.r.t "x"

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2)^{\frac{1}{2}}$$

$$= \frac{1}{2} (x^2 + y^2)^{\frac{1}{2}-1} \cdot (2x)$$

$$= \frac{1}{2} (x^2 + y^2)^{\frac{1}{2}-1} \cdot (2x)$$

$$\frac{\partial b}{\partial x} = (x^2 + y^2)^{-1/2} \cdot (x)$$

$$= \frac{x}{(x^2 + y^2)^{1/2}}$$

$$\frac{\partial b}{\partial x} = \frac{300}{(300^2 + 400^2)^{1/2}} \quad \Bigg| = 0.6 \quad \Bigg|$$

partially diff ① w.r.t "y"

$$\frac{\partial b}{\partial y} = \frac{\partial}{\partial y} (x^2 + y^2)^{1/2}$$

$$= \frac{1}{2} (x^2 + y^2)^{\frac{1}{2}-1} \cdot (2y)$$

$$= (x^2 + y^2)^{-1/2} \cdot y$$

$$= \frac{y}{(x^2 + y^2)^{1/2}}$$

= put values

$$= \frac{400}{(300^2 + 400^2)^{1/2}}$$

$$\Bigg| \frac{\partial b}{\partial y} = 0.8 \quad \Bigg|$$

Now Simplify:

• Chang in function

$$\Delta b = \frac{\partial b}{\partial x} \cdot dx + \frac{\partial b}{\partial y} \cdot dy$$

$$= (0.6) \cdot (-1) + (0.8) \cdot (-1)$$

$$\Delta b = -1.4$$

$$\therefore \sqrt{299^2 + 399^2} = \sqrt{x^2 + y^2} + \Delta b$$

$$= \sqrt{300^2 + 400^2} + (-1.4)$$

$$\sqrt{299^2 + 399^2} = 498.6 \quad \text{Ans:}$$

2) If $\theta = \arctan\left(\frac{y}{x}\right)$, use differentials to find an approximate value of θ when $x = 0.95$ and $y = 1.05$.

Sol:

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) \quad , \quad x = 0.95$$

$$y = 1.05$$

Most approach value of x and y is

$$0.95 \approx 1 = x$$

$$\theta(x, y) = \theta$$

$$1.05 \approx 1 = y$$

$$= \tan^{-1}\left(\frac{y}{x}\right) + \Delta\theta$$

$$\therefore \Delta\theta = \frac{\partial\theta}{\partial x} \cdot dx + \frac{\partial\theta}{\partial y} \cdot dy$$

Sol:

According to given statement :-

$$\theta = \tan^{-1}\left[\frac{y}{x}\right] \quad \text{--- (1)}$$

Let approach values are :-

$$1 + dx = 0.95$$

$$; \quad 1 + dy = 1.05$$

$$dx = 0.95 - 1$$

$$; \quad dy = 1.05 - 1$$

$$dx = -0.05$$

$$; \quad dy = 0.05$$

Partially diff (1) w.r.t "x" :-

$$\frac{\partial\theta}{\partial x} = \frac{\partial}{\partial x} \left[\tan^{-1}\left(\frac{y}{x}\right) \right]$$

$$= \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right)$$

$$= \frac{1}{1 + \frac{y^2}{x^2}} \cdot y \frac{\partial}{\partial x} \left(\frac{1}{x} \right)$$

$$= \frac{1}{x^2 + y^2} \cdot y \cdot \left(\frac{-1}{x^2} \right)$$

$$= \frac{\cancel{x^2}}{x^2 + y^2} \cdot \left(\frac{-y}{\cancel{x^2}} \right)$$

$$\frac{\partial \theta}{\partial x} = \frac{-y}{x^2 + y^2} = -$$

$$= \frac{-1}{(1)^2 + (1)^2}$$

$$\left| \frac{\partial \theta}{\partial x} = \frac{-1}{2} \right|$$

Partially diff θ w.r.t "y"

$$\frac{\partial \theta}{\partial y} = \frac{\partial}{\partial y} \left[\tan^{-1} \left(\frac{y}{x} \right) \right] \quad \therefore \frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2} \cdot \frac{d}{dx}(x)$$

$$= \frac{1}{1 + \left(\frac{y}{x} \right)^2} \cdot \frac{\partial}{\partial y} \left(\frac{y}{x} \right)$$

direct LiC: 19, x

$$= \frac{x^2}{x^2 + y^2} \cdot \frac{1}{x} \cdot \frac{\partial}{\partial y} (y)$$

$$= \frac{\cancel{x^2}}{x^2 + y^2} \cdot \frac{1}{\cancel{x}} \cdot (1)$$

$$\frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2}$$

$$= \frac{1}{1^2 + 1^2}$$

$$\left| \frac{\partial \theta}{\partial y} = \frac{1}{2} \right|$$

$$\therefore d\theta = \frac{\partial \theta}{\partial x} \cdot dx + \frac{\partial \theta}{\partial y} \cdot dy$$

$$= \frac{-1}{2} \cdot (-0.05) + \frac{1}{2} (0.05)$$

$$= 40 = 0.05$$

Now

$$\therefore \tan^{-1} \left(\frac{1.05}{0.95} \right) = \tan^{-1} \left(\frac{Y}{X} \right) + 40$$

$$= \tan^{-1} \left(\frac{1}{1} \right) + 0.05$$

$$= 45^\circ + 0.05$$

$$= \frac{45 \times \pi}{180} + 0.05$$

$$= 0.8354 \text{ rad Ans!}$$

- ③ If $u = \sqrt{x+2y}$ and x changes from 3 to 2.98 while y changes from 0.5 to 0.51, find an approximate value for the change in u .

Sol:

$$u = \sqrt{x+2y} \quad \text{--- ①}$$

$$\Rightarrow u = \sqrt{x+2y} \quad \text{--- ②}$$

Represented value

$$x = 2.98$$

$$y = 0.51$$

Approached value

$$X = 3$$

$$Y = 0.5$$

$$3 + dX = 2.98$$

$$dX = 2.98 - 3$$

$$dX = -0.02$$

;

$$0.5 + dY = 0.51$$

$$dY = 0.51 - 0.5$$

$$dY = 0.01$$

;

;

Partially diff ② w.r.t "x"

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \sqrt{x+2y}$$

$$\frac{\partial u}{\partial x}$$

$$= \frac{\partial}{\partial x} (x+2y)^{1/2}$$

$$= \frac{1}{2} (x+2y)^{\frac{1}{2}-1} \cdot (1)$$

Put values:

$$= \frac{1}{2} (3+2(0.5))^{-1/2}$$

$$\left| \frac{\partial u}{\partial x} = \frac{1}{4} \right|$$

partial diff w.r.t "y"

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x+2y)^{1/2}$$

$$= \frac{1}{2} (x+2y)^{\frac{1}{2}-1} \cdot (2)$$

$$= \frac{1}{2} (x+2y)^{-1/2} \cdot (2)$$

$$= (x+2y)^{-1/2}$$

put values

$$= (3+2(0.5))^{-1/2}$$

$$\left| \frac{\partial u}{\partial y} = \frac{1}{2} \right|$$

put all values in formula and get the value of change in u.

$$\Delta u = \frac{\partial u}{\partial x} \cdot dx + \frac{\partial u}{\partial y} \cdot dy$$

$$= \left(\frac{1}{4}\right) \cdot (-0.02) + \left(\frac{1}{2}\right) \cdot (0.01)$$

$$\left| \Delta u = 0 \right|$$

This show that, there is no change in given function. "U".

- (4) If $U = x^2 + y^2 + z^2 + xyz^3$ and x changes from 2 to 2.01, y changes from 1 to 1.02 and z changes from -1 to -0.99, find an approximate value for the change in U .
Sol:

$$U = x^2 + y^2 + z^2 + xyz^3 \quad \text{--- (1)}$$

$$U = X^2 + Y^2 + Z^2 + XYZ^3 \quad \text{--- (2)}$$

Represented values

$$x = 2.01$$

$$y = 1.02$$

$$z = -0.99$$

Approached values

$$X = 2$$

$$Y = 1$$

$$Z = -1$$

$$2 + dx = 2.01$$

$$dx = 2.01 - 2$$

$$dx = 0.01$$

$$1 + dy = 1.02$$

$$dy = 1.02 - 1$$

$$dy = 0.02$$

$$-1 + dz = -0.99$$

$$dz = -0.99 + 1$$

$$dz = 0.01$$

Partial diff w.r.t "x"

$$\frac{\partial U}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2 + z^2 + xyz^3)$$

$$= 2x + 0 + 0 + y^2 z^3$$

$$= 2x + y^2 z^3$$

$$= 2(2.01) + (1)^2 (-1)^3$$

$$\left| \frac{\partial U}{\partial x} = 3 \right|$$

Partial diff w.r.t "Y"

$$\frac{\partial U}{\partial Y} = \frac{\partial}{\partial Y} (x^2 + y^2 + z^2 + xy^2z^3)$$

$$= 0 + 2Y + 0 + 2XYZ^3$$

$$= 2Y + 2XYZ^3$$

$$= 2(1) + 2(2)(1)(-1)^3$$

$$\left| \frac{\partial U}{\partial Y} = -2 \right|$$

Partial diff w.r.t "Z"

$$\frac{\partial U}{\partial Z} = \frac{\partial}{\partial Z} (x^2 + y^2 + z^2 + xy^2z^3)$$

$$= 0 + 0 + 2Z + 3xy^2z^2$$

$$= 2Z + 3xy^2z^2$$

$$= 2(-1) + 3(2)(1)^2(-1)$$

$$= -2 + 6$$

$$\left| \frac{\partial U}{\partial Z} = 4 \right|$$

Now find change in "U"

$$\Delta U = \frac{\partial U}{\partial X} \cdot dX + \frac{\partial U}{\partial Y} \cdot dY + \frac{\partial U}{\partial Z} \cdot dZ$$

$$= 3(0.01) + (-2)(0.02) + 4(0.01)$$

$$\Delta U = 0.03$$

- ⑤ A rectangular plate expands in such a way that its length change from 10 to 10.03 and its breadth change from 8 to 8.02. Find an approximate value for the change in its area.
- Sol.

find change in area:

$$\Delta A = \frac{\partial A}{\partial X} \cdot dX + \frac{\partial A}{\partial Y} \cdot dY$$

Let;

length of rectangular plate = X

breadth of rectangular plate = Y

$$A = XY \quad \text{--- (1)}$$

Represented value

$$x = 10.03$$

$$y = 8.02$$

Approached value

$$X = 10$$

$$Y = 8$$

$$10 + dX = 10.03$$

$$dX = 10.03 - 10$$

$$dX = 0.03$$

$$; \quad 8 + dY = 8.02$$

$$; \quad dY = 8.02 - 8$$

$$; \quad dY = 0.02$$

Partial diff w.r.t "X"

$$\frac{\partial A}{\partial X} = \frac{\partial}{\partial X} (X \cdot Y)$$

$$= 1(Y)$$

$$\frac{\partial A}{\partial X} = Y$$

$$; Y = 8$$

$$\left| \begin{array}{c} \text{put values} \\ \frac{\partial A}{\partial X} = 8 \end{array} \right|$$

Partial diff w.r.t "Y"

$$\frac{\partial A}{\partial Y} = \frac{\partial}{\partial Y} (X \cdot Y)$$

$$= X(1)$$

$$= X$$

$$\therefore X = 10$$

put values

$$\left| \frac{\partial A}{\partial Y} = 10 \right|$$

Now put all values in formula:

$$\begin{aligned} \Delta A &= \frac{\partial A}{\partial X} \cdot dX + \frac{\partial A}{\partial Y} \cdot dY \\ &= 8(0.03) + 10(0.02) \end{aligned}$$

$$\left| \Delta A = 0.44 \right|$$

⑥

The lateral surface of a cone is computed from the formula $S = \pi r \sqrt{r^2 + h^2}$ where r is the radius of the base and h is the height. If r is calculated as 6 with an accuracy of 1% and h as 8 with an accuracy of 0.25%, with what accuracy will be the area S ?

Sol:

$$S = \pi r \sqrt{r^2 + h^2} \quad \text{--- (1)}$$

$$r = 6 \quad ; \quad h = 8$$

$$\text{change in } r = dr = 6 \times 1\%$$

$$dr = 0.06$$

$$\text{change in } h = dh = 8 \times 0.25\%$$

$$= dh = 0.02$$

∴ accuracy of $S = \frac{\Delta S}{S} \times 10$

∴ $\Delta S = \frac{\partial S}{\partial r} \cdot dr + \frac{\partial S}{\partial h} \cdot dh$

Now partially diff w.r.t "x"

$$\begin{aligned} \frac{\partial S}{\partial r} &= \frac{\partial S}{\partial} \left(\pi r \sqrt{r^2 + h^2} \right) \\ &= \frac{\partial S}{\partial} \left(\pi r (r^2 + h^2)^{1/2} \right) \\ &= \pi \left[r \cdot \frac{1}{2} (r^2 + h^2)^{1/2 - 1} \cdot (2r) + (1) \cdot \sqrt{r^2 + h^2} \right] \end{aligned}$$

$$= \pi \left[r \cdot (r^2 + h^2)^{-1/2} \cdot r \right] + 1 \cdot \sqrt{r^2 + h^2}$$

$$= \pi \left[\frac{r^2}{\sqrt{r^2 + h^2}} + \sqrt{r^2 + h^2} \right]$$

Put values

$$= \pi \left[\frac{6^2}{\sqrt{6^2 + 8^2}} + \sqrt{6^2 + 8^2} \right]$$

$$= \pi \left[\frac{36}{\sqrt{36 + 64}} + \sqrt{36 + 64} \right]$$

$$\left| \frac{\partial S}{\partial r} = 13.6\pi \right|$$

Partial

diff w.r.t "y"

$$\begin{aligned} \frac{\partial S}{\partial h} &= \frac{\partial S}{\partial} \left(\pi r (r^2 + h^2)^{1/2} \right) \\ &= \pi r \left[\frac{1}{2} (r^2 + h^2)^{1/2 - 1} \cdot (2h) \right] \end{aligned}$$

$$= \pi r \left[(r^2 + h^2)^{1/2} \cdot (h) \right]$$

$$= \frac{\pi r h}{\sqrt{r^2 + h^2}}$$

put values

$$= \frac{\pi \cdot (6)(8)}{\sqrt{6^2 + 8^2}}$$

$$= \frac{\pi \cdot 48}{\sqrt{36 + 64}}$$

$$\left| \frac{\partial S}{\partial h} = 4.8\pi \right|$$

Now put in formula.

$$\Delta S = \frac{\partial S}{\partial r} \cdot dr + \frac{\partial S}{\partial h} \cdot dh$$

$$= (13.6\pi)(0.06) + (4.8\pi)(0.02)$$

$$\Delta S = 0.912\pi$$

$$\text{Accuracy of } S = \frac{\Delta S}{S} \times 100$$

$$= \frac{0.912\pi}{60\pi} \times 100$$

$$\left| S = 1.52\% \right|$$

① The volume V of a rectangular parallelepiped having sides x, y and z is given by the formula $V = xyz$. If this solid is compressed from above so that z is decreased by 2% while x and y each is increased by 75% approximately, what percentage change will be in V ?
Sol:

$$V = xyz \quad \text{--- ①}$$

$$dx = +75\%$$

$$dy = +75\%$$

$$dz = -2\%$$

$$\therefore \% \text{ change in } V = \frac{\Delta V}{V} \times 100\%$$

Let

$$x = y = z = 1$$

Find $\rightarrow V, \Delta V$

put values of x, y, z in ①

$$V = (1)(1)(1)$$

$$V = 1$$

partial diff w.r.t "x"

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x} (xyz)$$

$$= yz \cdot \frac{\partial}{\partial x} (x)$$

$$= yz \cdot 1$$

put values $\therefore z = 1, y = 1$

$$= yz \cdot 1$$

$$= (1)(1) \cdot 1$$

$$\left| \frac{\partial V}{\partial x} = 1 \right|$$

① The volume V of a rectangular parallelepiped having sides x, y and z is given by the formula $V = xyz$. If this solid is compressed from above so that z is decreased by 2% while x and y each is increased by 75% approximately, what percentage change will be in V ?

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Find $\rightarrow V, \Delta V$

put values of x, y, z in ①

$$V = (1)(1)(1)$$

$$V = 1$$

partial diff w.r.t " x "

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x} (xyz)$$

$$= yz \cdot \frac{\partial}{\partial x} (x)$$

$$= yz \cdot 1$$

put values

$$\because z = 1, y = 1$$

$$= yz \cdot 1$$

$$= (1)(1) \cdot 1$$

$$\left| \frac{\partial V}{\partial x} = 1 \right|$$

Partial diff of w.r.t "y"

$$\frac{\partial V}{\partial y} = \frac{\partial V}{\partial} (xyz)$$

$$= xz \cdot \frac{\partial}{\partial y} (y)$$

$$= xz \cdot (1)$$

put values

$$= xz (1)$$

$$= (1)(1)(1)$$

$$x=1, z=1$$

$$\left| \frac{\partial V}{\partial y} = 1 \right|$$

Partial diff w.r.t "z"

$$\frac{\partial V}{\partial z} = \frac{\partial V}{\partial} (xyz)$$

$$= xy \cdot \frac{\partial}{\partial z} (z)$$

$$= xy \cdot (1)$$

put values

$$x=1, y=1$$

$$= xy \cdot (1)$$

$$= (1)(1) \cdot (1)$$

$$\left| \frac{\partial V}{\partial z} = 1 \right|$$

Now find change in V

$$\Delta V = \frac{\partial V}{\partial x} \cdot dx + \frac{\partial V}{\partial y} \cdot dy + \frac{\partial V}{\partial z} \cdot dz$$

$$= (1) \cdot (-0.75\%) + (1) \cdot (-0.75\%) + (1) \cdot (-2\%)$$

$$\Delta V = -0.005$$

Now

$$\% \text{ change in } V = \frac{\Delta V}{V} \times 100\%$$

$$= \frac{-0.005}{1} \times 100\%$$

$$= -0.5\%$$

⑧

A formula for the area Δ of a triangle is $\Delta = \frac{1}{2} ab \sin C$, where a, b are two adjacent side and C is the angle included.

Approximately what error is made in computing Δ if a is taken to be 9.1 instead of 9, b is taken to be 4.08 instead of 4 and C is take to be $30^\circ 3'$ instead of 30° ?

Sol:

$$\text{Area of } \Delta = \frac{1}{2} ab \sin C \quad \text{--- (1)}$$

$$a = 9 \quad ; \quad a + da = 9.1 \Rightarrow 9.1 - 9 \Rightarrow da = 0.1$$

$$b = 4 \quad ; \quad b + db = 4.08 \Rightarrow 4.08 - 4 \Rightarrow db = 0.08$$

$$C = 30^\circ \quad ; \quad 30^\circ + dC = 30^\circ 3' \Rightarrow 30^\circ 3' - 30^\circ \Rightarrow dC = 3'$$
$$dC = 0.000873 = \left(\frac{3}{60}\right)^\circ \times \left(\frac{\pi}{180}\right)$$

$$da = 0.1$$

$$db = 0.08$$

$$dC = 0.000873$$

find:

$$\text{Error in Area} = \frac{\Delta \text{Area}}{\text{Area}} \times 100\%$$

Now find Area:

put values of a, b & c in ① :-

$$\text{Area} = \frac{1}{2} (9)(4) \cdot \sin(30^\circ)$$

$$\text{Area} = 9$$

Taking partial diff ① w.r.t 'a' :-

$$\frac{\partial A}{\partial a} = \frac{\partial}{\partial a} \left(\frac{1}{2} ab \sin C \right)$$

$$= \frac{1}{2} b \sin C \cdot \frac{\partial a}{\partial a}$$

$$= \frac{1}{2} b \sin C \cdot (1)$$

$$= \frac{1}{2} b \sin C \cdot (1)$$

put values :-

$$= \frac{1}{2} (4) \sin(30^\circ) \cdot (1)$$

$$\left| \frac{\partial A}{\partial a} = 1 \right|$$

Partial diff ① w.r.t "b" :-

$$= \frac{1}{2} a \sin C \cdot \frac{\partial b}{\partial b}$$

$$= \frac{1}{2} a \sin C \cdot (1)$$

put values :-

$$= \frac{1}{2} (9) \sin(30^\circ) \cdot (1)$$

$$\left| \frac{\partial A}{\partial b} = 2.25 \right|$$

Partial diff ① w.r.t "C" :-

$$\frac{\partial A}{\partial C} = \frac{\partial}{\partial C} \left[\frac{1}{2} ab \sin C \right]$$

$$= \frac{1}{2} ab \cdot \frac{\partial}{\partial C} (\sin C)$$

$$= \frac{1}{2} ab \cdot \cos C$$

put values

$$= \frac{1}{2} (9)(4) \cdot \cos(30^\circ)$$

$$\left| \frac{\partial A}{\partial C} = 15.588 \right|$$

Now put all values in formula:

$$\Delta \text{Area} = \frac{\partial A}{\partial a} \cdot da + \frac{\partial A}{\partial b} \cdot db + \frac{\partial A}{\partial C} \cdot dC$$

$$= 1 \cdot (0.1) + 2.25(0.08) + 15.588(0.000873)$$

$$\left| \Delta \text{Area} = 0.294 \right|$$

Find Error:

$$\text{Error in Area} = \frac{\Delta \text{Area}}{\text{Area}} \times 100\%$$

$$= \frac{0.294}{9} \times 100\%$$

$$\left| \text{Error} = 3.2666\% \right|$$

Q

The dimensions of a box are measured to be 10 in, 12 in and 15 in. and measurement are correct to 0.02 in. Find the maximum error if the volume is calculated from the given measurements. Also find the percentage error.

Sol:

Let

$$x = 10 \text{ inch} \quad ; \quad y = 12 \text{ inch}$$

$$z = 15 \text{ inch}$$

$$dx = dy = dz = 0.02 \text{ inch}$$

$$V = xyz \quad \text{--- (1)}$$

$$\% \text{ error} = \frac{\Delta V}{V} \times 100\%$$

$$\Delta V = \frac{\partial V}{\partial x} \cdot dx + \frac{\partial V}{\partial y} \cdot dy + \frac{\partial V}{\partial z} \cdot dz$$

Now,

$$V = xyz$$

$$V = (10)(12)(15)$$

$$\text{Max } E = \left| V = 1800 \right| \text{ cubic inches.}$$

Partial diff (1) w.r.t "x" :-

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x} (xyz)$$

$$= yz \cdot \frac{\partial}{\partial x} (x)$$

$$= yz \cdot (1)$$

$$= yz \cdot (1) \quad \text{--- (2)}$$

put values in (2)

$$= (12)(15) \cdot 1$$

$$\left| \frac{\partial V}{\partial x} = 180 \right|$$

Partial diffb (i) w.r.t "y"

$$\frac{\partial V}{\partial y} = \frac{\partial}{\partial y} (xyz)$$

$$= xz \cdot \frac{\partial}{\partial y} (y)$$

$$= xz \cdot (1)$$

put values

$$= (10)(15) \cdot (1)$$

$$\left| \frac{\partial V}{\partial y} = 150 \right|$$

Partial diffb (i) w.r.t "z" :-

$$\frac{\partial V}{\partial z} = \frac{\partial}{\partial z} (xyz)$$

$$= xy \cdot \frac{\partial}{\partial z} (z)$$

$$= xy \cdot (1)$$

put values

$$= (10)(12) \cdot (1)$$

$$\left| \frac{\partial V}{\partial z} = 120 \right|$$

Now put in formula

$$\Delta V = \frac{\partial V}{\partial x} \cdot dx + \frac{\partial V}{\partial y} \cdot dy + \frac{\partial V}{\partial z} \cdot dz$$

$$AV = \frac{(180)(0.02) + 150(0.02) + 120(0.02)}{9}$$

Now,

$$\% \text{ error} = \frac{AV}{V} \times 100\%$$

$$= \frac{9}{1800} \times 100\%$$

$$\% \text{ error} = 0.5\%$$

⑩

Evaluate $\sin 29^\circ \cos 28^\circ \tan 44^\circ$
by using differentials.

Sol:

$$\sin 29^\circ ; \cos 28^\circ ; \tan 44^\circ$$

Let

$$x = 30^\circ ; y = 30^\circ ; z = 45^\circ$$

$$U = \sin x \cdot \cos y \cdot \tan z \quad \text{--- (1)}$$

$30^\circ + dx = 29^\circ$	$30^\circ + dy = 28^\circ$	$45^\circ + dz = 44^\circ$
$dx = 29^\circ - 30^\circ$	$dy = 28^\circ - 30^\circ$	$dz = 44^\circ - 45^\circ$
$dx = -1^\circ$	$dy = -2^\circ$	$= -1^\circ$
$= -1 \times \frac{\pi}{180}$	$= -2 \times \frac{\pi}{180}$	$= -1 \times \frac{\pi}{180}$

$$dx = -0.0175 \quad dy = -0.0349 \quad dz = -0.0175$$

Find:-

$$\text{Approximate} = U + \Delta U$$

put all values in (1)

$$U = \sin x \cdot \cos y \cdot \tan z$$

$$= (\sin 30^\circ)(\cos 30^\circ)(\tan 45^\circ)$$

$$U = 0.433$$

Partial diff ① wrt "x"

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x} (\sin x \cdot \cos y \cdot \tan z)$$

$$= \cos y \cdot \tan z \cdot \frac{\partial}{\partial x} (\sin x)$$

$$= \cos y \cdot \tan z \cdot \cos x$$

put values

$$= \cos(30)^\circ \cdot \tan(45)^\circ \cdot \cos(30)^\circ$$

$$\left| \frac{\partial V}{\partial x} = 0.75 \right|$$

Partial diff ① wrt "y"

$$\frac{\partial V}{\partial y} = \frac{\partial}{\partial y} (\sin x \cdot \cos y \cdot \tan z)$$

$$= \sin x \cdot \tan z \cdot \frac{\partial}{\partial y} (\cos y)$$

$$= \sin x \cdot \tan z \cdot -\sin y$$

put values

$$= \sin(30)^\circ \cdot \tan(45)^\circ \cdot -\sin(30)^\circ$$

$$\left| \frac{\partial V}{\partial y} = -0.25 \right|$$

Partial diff ① with "z"

$$\frac{\partial V}{\partial z} = \frac{\partial}{\partial z} (\sin x \cdot \cos y \cdot \tan z)$$

$$= \sin x \cdot \cos y \cdot \frac{\partial}{\partial z} (\tan z)$$

$$= \sin x \cdot \cos y \cdot \sec^2 z$$

put values

$$= \sin(30)^\circ \cdot \cos(30)^\circ \cdot \sec^2(45)^\circ$$

$$\left| \frac{\partial V}{\partial z} = 0.866 \right|$$

Now put in formula

$$\begin{aligned}\Delta U &= \frac{\partial U}{\partial x} \cdot dx + \frac{\partial U}{\partial y} \cdot dy + \frac{\partial U}{\partial z} \cdot dz \\ &= 0.75(-0.0175) + (0.25)(-0.0349) + 0.856(-0.0175) \\ | \Delta U &= -0.019555 | \end{aligned}$$

$$\begin{aligned}\sin 29^\circ \cdot \cos 28^\circ \cdot \tan 44^\circ &= U + \Delta U \\ &= 0.433 + (-0.019555) \\ &= 0.413445\end{aligned}$$